Strand 1: Number

Sub-strand 1: Number and Numeration Systems

Lesson 2: Real Number System, Sets, and Venn Diagrams

1. Curriculum Standard and Overview

- **Strand**: 1 — Number
- **Sub-strand**: 1 — Number and Numeration Systems
- **Content Standard**: B9.1.1.2 — Understand the structure of the real number system and use set ideas to solve problems.
- **Indicators**:
- **B9.1.1.2.1** — Solve problems on relationships between members of the rational number system.
- **B9.1.1.2.2** — Apply the concept of sets to solve problems on relationships between members of the number system and in real-life surveys.
- **Scope**:
  1. The number family tree: Natural → Whole → Integers → Rational → Real.
  2. Deciding which set a number belongs to.
  3. Set language: elements, universal set, complement, union, intersection.
  4. Drawing and reading 2-set and 3-set Venn diagrams.
  5. Survey word problems (the classic "in a market of …" type).
- **Mock-paper links**: Paper 1 Q15 (largest set); Paper 2 Q5(d) (fishmongers Venn diagram).

2. Preamble — Numbers Live in Families

You already know counting numbers: 1, 2, 3, … But maths also needs zero, negatives, fractions, and numbers like √2 that never end.

Mathematicians **group numbers into families** (called **sets**) so we can talk about "what kind of number is this?" precisely.

Key word — SET. *A set is simply a* ***collection of well-defined objects****, usually written inside curly brackets: {1, 2, 3}.* ***Key word — ELEMENT.*** *Each member of a set is an element. We write ∈ for "is an element of" and ∉ for "is not".*

3. The Number Family Tree

3.1 The five big families

Symbol	Name	What it contains	Examples
■	Natural (counting) numbers	1, 2, 3, 4, …	7, 100
■	Whole numbers	Natural numbers + 0	0, 5
■	Integers	Whole numbers + negatives	–9, 0, 12
■	Rational numbers	Any number you can write as p/q (p, q integers, q ≠ 0). Includes terminating and recurring decimals	3/4, –2, 0.25

■′	Irrational numbers	Numbers that **cannot** be written as p/q; decimals go on forever without repeating	√2, π

Together, ■ and ■′ make the **Real numbers** (■).

3.2 Nesting picture (each family is inside the one after it)

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ REAL NUMBERS ■
■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
■ ■ RATIONAL (■) ■ ■
■ ■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ ■
■ ■ ■ INTEGERS (■) ■ ■ ■
■ ■ ■ ■■■■■■■■■■■■■■■■■■■■■■■■■ ■ ■ ■
■ ■ ■ ■ WHOLE (■) ■ ■ ■ ■
■ ■ ■ ■ ■■■■■■■■■■■■■■■■■ ■ ■ ■ ■
■ ■ ■ ■ ■ NATURAL (■) ■ ■ ■ ■ ■
■ ■ ■ ■ ■ 1, 2, 3, … ■ ■ ■ ■ ■
■ ■ ■ ■ ■■■■■■■■■■■■■■■■■ ■ ■ ■ ■
■ ■ ■ ■ + 0 ■ ■ ■ ■
■ ■ ■ ■■■■■■■■■■■■■■■■■■■■■■■■■ ■ ■ ■
■ ■ ■ + −1, −2, −3, … ■ ■ ■
■ ■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ ■
■ ■ + ½, 0.25, −3.7, … ■ ■
■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
■ Irrationals (■′): √2, π, … ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

3.3 How to classify a number quickly

Ask these questions **in order**:

1. Is it a negative whole number? → Integer only (not whole/natural).
2. Is it a positive whole number (or zero)? → Natural and whole (and integer and rational).
3. Can it be written exactly as a fraction p/q? → Rational.
4. Does its decimal go on forever with no repeating pattern? → Irrational.

Examples:

- 5 → Natural, Whole, Integer, Rational.
- 0 → Whole, Integer, Rational (not natural).
- –8 → Integer, Rational.
- 0.75 = 3/4 → Rational.
- 1.333… = 4/3 → Rational (recurring).
- √2 ≈ 1.414213… → Irrational.

4. Set Notation You Must Know

Symbol	Meaning	Example
∈	is an element of	5 ∈ {1, 3, 5}
∉	is not an element of	4 ∉ {1, 3, 5}
∪	union (everything in either set)	{1,2} ∪ {2,3} = {1,2,3}
∩	intersection (only what is in both)	{1,2} ∩ {2,3} = {2}
A′ or A■	complement of A (not in A, but in U)	If U = {1..6}, A = {1,2}, A′ = {3,4,5,6}
U or ξ	universal set (everything we are talking about)	
n(A)	number of elements in A (the **count**)	n({2,4,6}) = 3

⊂	is a subset of	■ ⊂ ■
∅ or {}	empty set	

5. Venn Diagrams

5.1 What a Venn diagram does

A Venn diagram is a **picture of sets**. A rectangle is the **universal set**. Each circle is one set. The overlapping region is the **intersection**.

5.2 Two-set Venn diagram layout

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Universal set U ■
■ ■■■■■■■■■■■■ ■■■■■■■■■■■■ ■
■ ■ only A ■ A ■ only B ■ ■
■ ■ ■ ∩ ■ ■ ■
■ ■ ■ B ■ ■ ■
■ ■■■■■■■■■■■■ ■■■■■■■■■■■■ ■
■ outside both = (A∪B)′ ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

5.3 The golden counting rule (2 sets)

For two sets A and B inside a universal set of size n(U):

`n(A \cup B) = n(A) + n(B) - n(A \cap B)`

We subtract the overlap once because adding n(A) and n(B) **counts the overlap twice**.

Rearranged:

`n(A only) = n(A) - n(A \cap B)` `n(B only) = n(B) - n(A \cap B)` `n(neither) = n(U) - n(A \cup B)`

Always fill a Venn from the middle (the overlap) outwards.

6. Twenty Fully Worked Examples

Example 1 — Classify 7

7 is a counting number → **Natural, Whole, Integer, Rational**.

Example 2 — Classify 0

0 is not a counting number → **Whole, Integer, Rational** (not natural).

Example 3 — Classify –12

Integer, Rational (also rational because –12 = –12/1).

Example 4 — Is 0.25 rational?

`0.25 = (1)/(4)`, so **yes — Rational**.

Example 5 — Is √9 rational?

`√(9) = 3`, which is a natural number → **Rational**. *(Careful: the square root symbol does not automatically mean irrational.)*

Example 6 — Is √7 rational?

7 is not a perfect square, so √7 = 2.6457… (no repeating block) → **Irrational**.

Example 7 — Largest of these sets

Which is the largest set: {whole numbers}, {composite numbers}, {natural numbers}, {integers}?

- Natural = {1,2,3,…}
- Whole = {0,1,2,3,…}
- Integers = {…,–2,–1,0,1,2,…}
- Composite = {4,6,8,9,10,…} (smallest family here)

Answer: {integers} (Mock 2026 P1 Q15 — option D).

Example 8 — Smallest of those sets

From the list above, the **composite numbers** form the smallest set.

Example 9 — Simple union

If A = {1, 2, 3} and B = {3, 4, 5}, find A ∪ B. A ∪ B = {1, 2, 3, 4, 5}.

Example 10 — Simple intersection

Using the same A and B: A ∩ B = {3}.

Example 11 — Complement

U = {1,2,3,4,5,6}, A = {1,2}. Find A′. A′ = {3, 4, 5, 6}.

Example 12 — Two-set counts

In a class of 40, 22 play football, 18 play hockey, 9 play both. How many play neither?

1. `n(F \cup H) = 22 + 18 - 9 = 31`.
2. Neither = `40 - 31 = 9`.

Answer: 9 students.

Example 13 — Fill the Venn from Example 12

Football only = `22 - 9 = 13`. Hockey only = `18 - 9 = 9`. Both = `9`. Neither = `9`. Check: `13 + 9 + 9 + 9 = 40` ✓

Example 14 — Mock Paper 2 Q5(d): fishmongers

In a market of 36 fishmongers, 28 sell mackerel (M), 18 sell tilapia (T), 13 sell both.

(i) Venn diagram:

- Both: 13
- Mackerel only: `28 - 13 = 15`
- Tilapia only: `18 - 13 = 5`
- Outside both: `36 - (15+13+5) = 3`

(ii) Not selling any of the two: **3**.

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ U (36) ■
■ ■■■■■■■■■■■ ■■■■■■■■■■■ ■
■ ■ 15 ■ 13 ■ 5 ■ ■
■ ■ mackerel■both ■ tilapia ■ ■
■ ■ only ■ ■ only ■ ■
■ ■■■■■■■■■■■ ■■■■■■■■■■■ ■
■ 3 ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

Example 15 — Finding n(A) from the picture

From Example 14, how many sell mackerel? Mackerel only + both = `15 + 13 = 28` ✓ (matches the question).

Example 16 — Three-set introduction

If we add a third set C (say, those who sell crabs), we draw a third circle. The centre region is $A \cap B \cap C$ (all three). We still fill the **innermost** regions first.

Example 17 — Subset

True or false: ■ ⊂ ■? True — every natural number is an integer.

Example 18 — True/False on membership

Is $-3 \in$ {whole numbers}? No → **False** (whole numbers start at 0).

Example 19 — Word → set

"The set of even numbers between 1 and 11." = {2, 4, 6, 8, 10}.

Example 20 — Survey with percentages

In a survey of 200 people, 120 like tea, 90 like coffee, and 40 like both. How many like neither?

`n(T \cup C) = 120 + 90 - 40 = 170`. Neither = `200 - 170 = 30`.

Answer: 30 people.

7. Comprehensive Exercise Sets (40 Questions)

*All authentic BECE/Mock questions for this topic are **included in full**. Generated questions top up to the 40-quota. Spiral review ≈ 15%.*

Exercise 1 — Identifying number families

1. **[Mock 2026 P1 Q15]** Which of the following sets is the largest?

A. {whole numbers} B. {composite numbers} C. {natural numbers} D. {integers}

2. Which of these sets is the **smallest**?

A. {whole numbers} B. {composite numbers} C. {natural numbers} D. {integers}

3. Classify 5 under every family that fits (■, ■, ■, ■).
4. Classify –3 under every family that fits.
5. Classify 0 under every family that fits.

Exercise 2 — Rational vs irrational

1. Is 0.25 rational? Give a reason.

2. Is $\sqrt{9}$ rational? Give a reason.
3. Is $\sqrt{7}$ rational or irrational?
4. Write 0.333… as a fraction; hence show it is rational.
5. **Spiral review:** Write 47 300 in standard form (Lesson 1 revision).

Exercise 3 — Set notation

1. If $A = \{1, 2, 3\}$, state whether $2 \in A$.
2. If $A = \{1, 2, 3\}$, state whether $4 \in A$.
3. $U = \{1,2,3,4,5,6\}$, $A = \{1,2\}$. Find A'.
4. Let $X = \{2,4,6\}$, $Y = \{4,6,8\}$. Find $X \cap Y$.
5. Let $X = \{2,4,6\}$, $Y = \{4,6,8\}$. Find $X \cup Y$.

Exercise 4 — Counting elements

1. Find $n(\{3, 5, 7, 9\})$.
2. Find $n(\varnothing)$.
3. If $n(A) = 7$ and $n(B) = 5$ and $A \cap B = \{2\}$, find $n(A \cup B)$.
4. **Spiral review:** Evaluate $36 \div (4 + 2) \times 5$ (BODMAS).
5. **Spiral review:** Round 0.006 845 to 2 s.f.

Exercise 5 — Two-set Venn diagrams (counts)

1. In a class of 40, 22 play football, 18 play hockey, 9 play both. How many play neither?
2. For the same class, how many play football only?
3. For the same class, how many play hockey only?
4. In a group of 50, 30 drink tea, 25 drink coffee, 10 drink both. How many drink neither?
5. In a group of 50, 30 drink tea, 25 drink coffee, 10 drink both. How many drink tea only?

Exercise 6 — Survey problems (the mock style)

1. **[Mock 2026 P2 Q5(d)]** In a market of 36 fishmongers, 28 sell mackerel, 18 sell tilapia and 13 of them sell both.

(i) Represent this information on a Venn diagram. (ii) How many in the market are not selling any of the two fishes?

2. In a survey of 200 people, 120 like tea, 90 like coffee, and 40 like both. How many like neither?
3. In a school of 180 pupils, 100 study French, 80 study Twi, and 30 study both. How many study neither?
4. From Question 3, how many study French only?
5. **Spiral review:** Convert 31 days into minutes and write the answer in standard form.

Exercise 7 — Subsets and true/false

1. True or false: ■ $\subset$ ■?
2. True or false: $-3 \in$ {whole numbers}?
3. Write "the set of even numbers between 1 and 11" in curly-bracket notation.
4. List all elements of $\{x : x \text{ is a factor of } 12\}$.
5. **Spiral review:** Write 5 040 007 in words.

Exercise 8 — Mixed and challenge

1. If $U = \{1,2,3,4,5,6,7,8,9,10\}$, $A = \{2,4,6,8,10\}$, $B = \{3,6,9\}$, find

(i) $A \cap B$ (ii) $A \cup B$ (iii) A'.

2. In a class of 35, every pupil studies at least one of Maths (M) and Science (S). 20 study M, 18 study S, and 8 study both. How many study Maths only?
3. A Venn diagram shows $n(U) = 60$, $n(A) = 25$, $n(B) = 30$, $n(A \cup B) = 45$. Find $n(A \cap B)$.
4. **Spiral review:** Arrange in ascending order: 3 050 082; 3 005 820; 3 500 082; 3 050 802.
5. **Spiral review:** Find the value of the digit 6 in 9 463 285.

8. Full Step-by-Step Worked Solutions

Solutions to Exercise 1

1. Integers contain negatives, zero, and all whole numbers — the biggest family listed. **Answer: D.**

2. Composite numbers (4, 6, 8, 9, …) are the most restricted. **Answer: B.**

3. 5 is a counting number, so **5** ∈ **■, ■, ■, ■**.

4. –3 is not counting/whole → **–3** ∈ **■, ■** only.

5. 0 is whole but not natural → **0** ∈ **■, ■, ■**.

Solutions to Exercise 2

1. `0.25 = (1)/(4)` (p/q form) → **Rational**.

2. √`(9) = 3 \in \mathbb{N}` → **Rational**.

3. √7 ≈ 2.645751… with no repeating block → **Irrational**.

4. Let `x = 0.333\ldots`; then `10x = 3.333\ldots`; subtract: `9x = 3 \Rightarrow x = (1)/(3)` → **Rational**.

5. `47 300 = 4.73 × 10^4`.

Solutions to Exercise 3

1. 2 is listed in A → **True**.

2. 4 is not listed in A → **False**.

3. A′ = {3, 4, 5, 6}.

4. X ∩ Y = {4, 6}.

5. X ∪ Y = {2, 4, 6, 8}.

Solutions to Exercise 4

1. Count the elements: **4**.

2. The empty set has **0** elements.

3. `n(A \cup B) = 7 + 5 - 1 = 11`.

4. `36 ÷ 6 = 6`; `6 × 5 = 30` → **30**.

5. 0.006 845 → 2 s.f. → **0.0068**.

Solutions to Exercise 5

1. `n(F \cup H) = 22 + 18 - 9 = 31`; neither = `40 - 31 = 9`.

2. Football only = `22 - 9 = 13`.

3. Hockey only = `18 - 9 = 9`.

4. `n(T \cup C) = 30 + 25 - 10 = 45`; neither = `50 - 45 = 5`.

5. Tea only = `30 - 10 = 20`.

Solutions to Exercise 6

1. **(i)** Overlap 13; M only = `28-13 = 15`; T only = `18-13 = 5`. **(ii)** Outside = `36 - (15+13+5) = 3`.

2. `120 + 90 - 40 = 170`; neither = `200 - 170 = 30`.

3. `100 + 80 - 30 = 150`; neither = `180 - 150 = 30`.

4. French only = `100 - 30 = 70`.

5. `31 × 1440 = 44 640 = 4.464 × 10^4` minutes.

Solutions to Exercise 7

1. Every natural number is an integer → **True**.

2. Whole numbers are 0, 1, 2, … → **False**.

3. {2, 4, 6, 8, 10}.

4. Factors of 12: 1, 2, 3, 4, 6, 12 → {1, 2, 3, 4, 6, 12}.

5. 5 040 007 = **five million, forty thousand, and seven**.

Solutions to Exercise 8

1. (i) $A \cap B = \{6\}$ (ii) $A \cup B = \{2, 3, 4, 6, 8, 9, 10\}$ (iii) $A' = \{1, 3, 5, 7, 9\}$

2. Maths only = `20 - 8 = 12`.

3. `n(A \cap B) = 25 + 30 - 45 = 10`.

4. 3 005 820 < 3 050 082 < 3 050 802 < 3 500 082.

5. Digit 6 is in the ten-thousands place → **60 000**.

9. Authentic BECE & Mock Questions Used

Bank / source	Question	Answer
Mock 2026 P1 Q15	Largest set?	D — {integers}
Mock 2026 P2 Q5(d)	Fishmongers Venn: 36 total, 28 M, 18 T, 13 both	Neither = 3

10. Lesson Summary

Skill	Rule
Classify numbers	Ask: integer? whole? fraction? non-repeating decimal?
Union	Everything in either set
Intersection	Only the overlap
Complement	Inside U but outside the set
Two-set count	$n(A\cup B) = n(A) + n(B) - n(A\cap B)$
Venn strategy	**Fill from the middle outwards**

Lesson prepared under `Grade9-Maths/Rules.md` *— plain English, British notation, full workings required.*